POSITION PAPER · ENGINEERING STANDARDS · AI SYSTEMS

The AI P&ID

A Notation Standard for Documenting Consequential AI Systems

Jonathan Horvat

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ABSTRACT

Process engineering documents complex systems using Piping and Instrumentation Diagrams — drawings that show not just topology but behavior, failure modes, control loops, instrument tags, and version history. These properties are what make formal safety analysis possible. AI system architecture is currently documented with block diagrams that show topology and nothing else. As AI systems become more physically consequential, block diagrams are insufficient. This paper proposes an AI P&ID notation standard — a symbol library, tag numbering convention, and drawing format adapted from ANSI/ISA-5.1-2024 and ISO 10628 — that gives AI systems the documentation discipline that process engineering developed over seventy years. The standard is demonstrated on the broadcast sync architecture from the companion paper series, producing the first AI P&ID in the proposed notation.

> *This paper is a companion to:* Horvat, J. (2026). *Intelligence at the Edge of the Cloud: A Local Architecture for the AI Era.* Public domain.

SECTION I

Why Block Diagrams Are Not Enough

A process engineer handed a block diagram of a chemical plant would not be able to conduct a HAZOP, write an operating procedure, specify a control system, or approve the design for construction. A block diagram shows components and connections. It does not show what flows between components, what data types those connections carry, how each component fails, what instruments monitor what parameters, what interlocks prevent unsafe states, or what version of the design is being reviewed. It is a sketch, not a document.

AI system architecture documentation in 2026 is almost entirely block diagrams. Boxes connected by arrows, sometimes with labels. When teams use more sophisticated tools they produce UML diagrams or data flow diagrams — both of which are improvements, but neither of which captures the properties

required for safety-critical system documentation. None of them show failure modes. None of them show what happens when a connection is lost. None of them have a tag numbering system that connects the diagram to a HAZOP register. None of them have formal version control in the engineering sense.

This is not a criticism of the people producing these diagrams. There is no standard. There are no conventions. Nobody has defined what a complete AI system diagram should contain. The result is that every team invents its own notation, diagrams are not comparable across organizations, and the formal safety analysis that requires a precise drawing cannot be done.

> *A block diagram tells you what a system is. A P&ID tells you how it behaves, how it fails, and who is responsible for each part of it. Consequential AI systems need the second kind of document.*

BLOCK DIAGRAM VS AI P&ID — PROPERTIES COMPARISON

PROPERTY	BLOCK DIAGRAM	AI P&ID (PROPOSED)
Component topology	✓ Shows components and connections	✓ Shows components and connections
Data types on connections	✗ Not specified	✓ Data type, size, and direction annotated on every line
Failure modes	✗ Not shown	✓ Every component shows fail-safe, fail-operational, or fail-stop
Instrument tags	✗ No tagging convention	✓ ISA-adapted tag numbering for every instrument and control loop
Control loops	✗ Not represented	✓ Monitoring instruments, controllers, and final control elements shown
Security boundaries	✗ Implied at best	✓ Enclave boundaries shown as defined zones with access annotations
Version control	✗ Informal or absent	✓ Revision block, date, change history, management of change required
HAZOP support	✗ Insufficient for node analysis	✓ Node boundaries defined; parameters readable from drawing
Legal standing	✗ Informal communication tool	✓ Controlled document with formal change management

→ [1, 2]

SECTION II

What a P&ID Actually Contains

A Piping and Instrumentation Diagram, as defined by ANSI/ISA-5.1-2024 and ISO 10628, is a detailed drawing of a process system showing all equipment, piping, instrumentation, and control systems with sufficient detail to support design, construction, operation, and safety analysis. It is not a schematic. It is not a concept diagram. It is an engineering document.

The properties that make a P&ID useful for safety analysis are precisely defined by the standards. Equipment is shown with its tag number, size, material of construction, and design conditions. Pipe lines are shown with their size, schedule, material, insulation, and direction of flow. Instruments are shown with tag numbers following the ISA letter-based convention — a pressure indicator is PI, a flow controller is FC, a temperature transmitter is TT. Each tag includes a loop number that connects it to the instrument data sheet and the control system documentation.

Most importantly for safety analysis, every valve shows its failure mode — fail closed (FC), fail open (FO), or fail in last position (FL). Every instrument shows its action on signal failure. Every safety interlock is shown with its trip setpoint. Every relief device is shown with its set pressure. The drawing contains enough information to ask, for every component: what happens when this fails, and does the system respond safely?

The drawing also has a title block with revision history. Every change to a P&ID triggers a management of change process. The revision number, date, description of change, and responsible engineer are recorded. You cannot change a P&ID informally. The version history is part of the document.

→ [1, 2, 3]

SECTION III

The AI P&ID Symbol Library

The following symbol library defines the graphical elements required to document AI systems in the proposed notation. Symbols follow the ISA convention of using circles for instruments, boxes for equipment, and lines for connections, adapted for AI-specific element types. Where no process equivalent exists, new symbols are defined with rationale.

AI P&ID SYMBOL LIBRARY — PROPOSED V1.0 ADAPTED FROM ANSI/ISA-5.1-2024 · ISO 10628

001

SYMBOL	TAG	ELEMENT NAME	DESCRIPTION	FAILURE MODE ANNOTATION	ISA/PROCESS ANALOG
CATEGORY 01 · MODELS — PRIMARY PROCESSING ELEMENTS					
WM	WM-XXX	World Model	Sparse expert inference model. Pull-only. Receives weight syncs. Responds to queries from augmented personal model only.	FL — Fail last known weights. Continues serving from current weights if sync unavailable.	Process vessel / reactor — primary transformation element
APM	APM-XXX	Augmented Personal Model	On-device model with frozen base, personal adapters, and interpreter head. Inside secure enclave. Primary user-facing element.	FL — Fail last state. Responds from personal adapters if world model unavailable.	Process vessel with internal components — complex transformation

					element
IH	IH-XXX	Interpreter Head	Fixed non-trainable cross-attention module. Frozen at compile time. Translates between world model and personal model embedding spaces. Internal to APM.	FO — Fail open to personal model only. If cross-attention fails, synthesize from personal adapters without world model input.	Heat exchanger / translator — embedded transformation element
CATEGORY 02 · ADAPTERS — TRAINABLE PARAMETER SETS					
PA	PA-XXX	Personal Adapter Set	Low-rank adapter layers encoding user patterns. Trainable. Updated by behavioral observation only. Never transmitted. Inside secure enclave.	FL — Fail last state. Adapter weights frozen if training signal unavailable. System continues at current calibration level.	Control valve with positioner — trainable final control element
CATEGORY 03 · INSTRUMENTS — MONITORING AND MEASUREMENT ELEMENTS					
CI 001	CI-XXX	Coverage Indicator	Monitors and displays current domain coverage score. Reads router activation confidence, expert freshness, and domain relevance. Surfaces signal to user interface.	FL — Fail to last known coverage state. If coverage computation fails, display last valid score with staleness timestamp.	Level indicator / analyzer — passive readout instrument (ISA circle symbol)
ST 001	ST-XXX	Sync Transmitter	Monitors sync channel status — last sync timestamp, sync success/failure, bytes received. Transmits status signal to coverage indicator and system monitor.	FL — Fail last state. If sync monitoring fails, retain last known sync timestamp.	Flow transmitter / status transmitter — signal originating instrument (ISA circle)
MA	MA-XXX	Model Analyzer	Monitors world model version, expert inventory, and calibration state. Analogous to a process analyzer — reads composition of active expert set.	FL — Fail last inventory. If model analyzer fails, retain last known expert inventory list.	Composition analyzer — quality measurement instrument (ISA circle with A)
CATEGORY 04 · CONNECTIONS — DATA FLOW LINES					
	—	Primary Data Flow	Main inference data path. Annotated with data type, approximate size, and direction. Heavy solid line.	*N/A*	Main process line — primary fluid flow
	—	Signal / Control Line	Instrument signal, control output, or monitoring connection. Dashed line. Used for coverage score output, monitoring signals, and control loops.	*N/A*	Instrument signal line — electrical or pneumatic signal
	—	Sync Channel	Weight broadcast channel from sync server to device. One-directional. Annotated with sync frequency, payload size, and anonymization method.	*N/A — channel loss results in FL behavior on world model*	Utility line — secondary service connection
CATEGORY 05 · BOUNDARIES — SECURITY AND SYSTEM ZONES					
ENCLAVE	SE-XXX	Secure Enclave Boundary	Hardware secure enclave boundary. Dashed red rectangle. All elements inside this boundary are hardware-isolated. No data crosses boundary without explicit interface definition.	FC — Fail closed. If enclave is compromised, all contained elements cease operation. No degraded operation outside enclave boundary.	Battery limit / system boundary — physical plant boundary
DEVICE	DB-XXX	Device Boundary	Physical device boundary. Light dashed rectangle. Defines what is on-device vs off-device. All primary inference operations should be within this boundary.	*N/A — boundary definition only*	Equipment boundary — physical equipment limit

The symbol library is not exhaustive. It defines the minimum set required to document the broadcast sync architecture. A full standard would include additional categories for attention mechanisms, embedding

layers, training loops, and deployment infrastructure. The symbol set should be extended by the standards body that adopts it, following the ISA model of periodic revision with industry consensus.

→ [1, 2, 4]

SECTION IV

Tag Numbering Convention

The ISA tag numbering system uses a letter-based convention where the first letter identifies the measured variable, subsequent letters identify the function, and a loop number makes each tag unique. FIC045 is the Flow Indicating Controller in loop 045. The convention has been in use since 1949 and is universally understood by process engineers.

The proposed AI P&ID tag convention adapts this structure directly. The first letter identifies the AI element type, the second identifies its function, and a sequential number makes it unique within the system.

AI P&ID TAG CONVENTION — ADAPTED FROM ANSI/ISA-5.1 LETTER-BASED IDENTIFICATION

[Type] [Function] – [Loop Number]

LETTER	TYPE / FIRST POSITION	FUNCTION / SECOND POSITION	EXAMPLE TAG
M	Model	—	WM-001 (World Model 001)
A	Adapter	—	PA-001 (Personal Adapter 001)
I	Interpreter	I (Indicator)	IH-001 (Interpreter Head 001)
C	Coverage	I (Indicator) · T (Transmitter) · C (Controller)	CI-001 (Coverage Indicator 001)
S	Sync	T (Transmitter) · V (Valve/Gate) · A (Alarm)	ST-001 (Sync Transmitter 001)
W	Weight	T (Transmitter) · I (Indicator)	WT-001 (Weight Transmitter 001)
E	Enclave	B (Boundary)	SE-001 (Secure Enclave 001)
D	Device	B (Boundary)	DB-001 (Device Boundary 001)
U	User Interface	I (Interface)	UI-001 (User Interface 001)
X	External System	I (Interface) · S (Server)	XS-001 (External Sync Server 001)

EXAMPLE TAG READS

WM-001	World Model, instance 001
APM-001	Augmented Personal Model, instance 001
CI-001	Coverage Indicator, loop 001 — reads coverage score and displays to user

CI-001	Coverage indicator, loop 001 — reads coverage score and displays to user
ST-001	Sync Transmitter, loop 001 — monitors sync channel and transmits status
SE-001	Secure Enclave boundary, instance 001 — hardware isolation zone containing APM-001 and PA-001
SA-001	Sync Alarm, loop 001 — activates when sync has not succeeded within defined interval

The tag numbering convention enables the AI P&ID to connect to a HAZOP register exactly as a process P&ID does. Every row in the HAZOP register references a tag. Every action item in the register references a tag. When the register says "SA-001 — Sync Alarm does not activate when sync interval exceeded," both the register and the drawing refer to the same uniquely identified element. The documentation system is self-consistent.

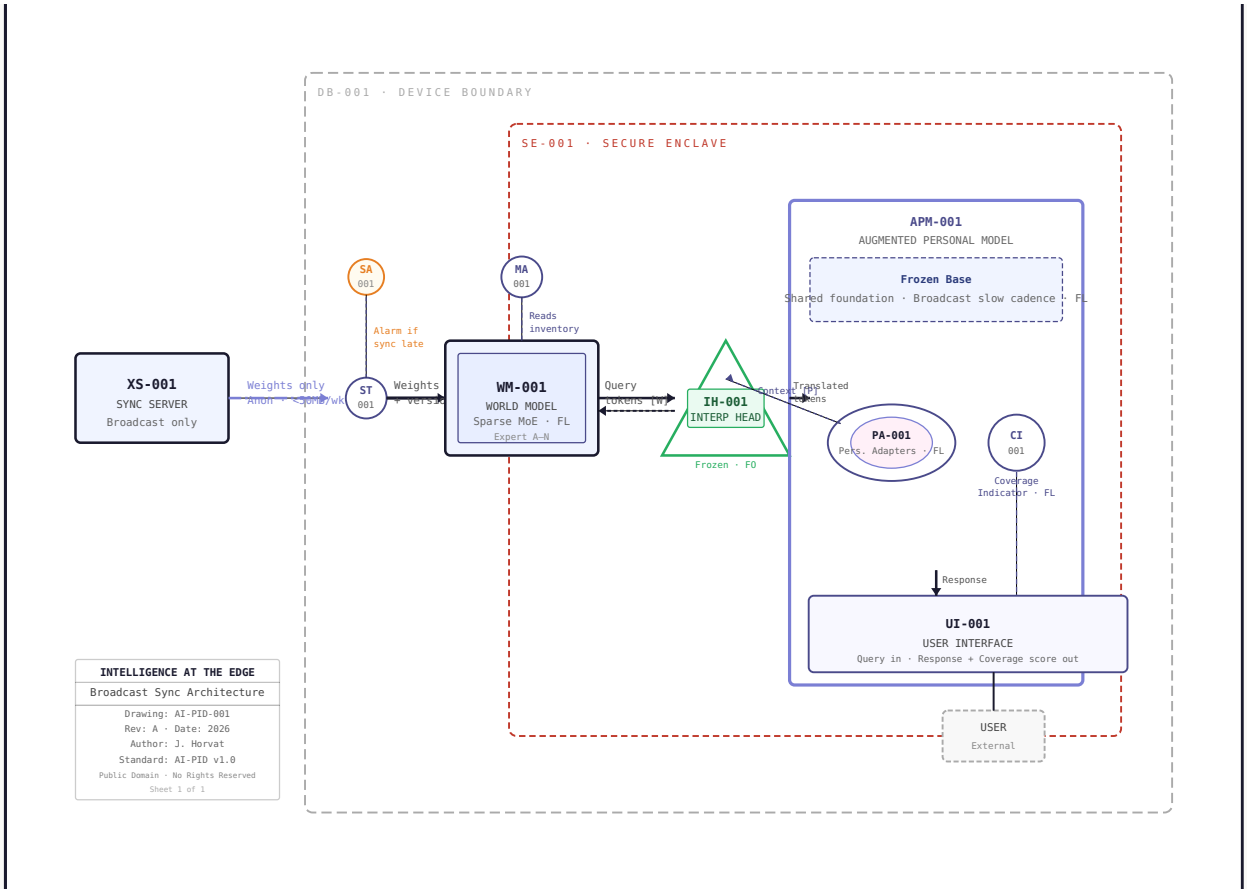
→ [1, 3]

SECTION V

The Demonstration Drawing

The following AI P&ID documents the broadcast sync architecture from *Intelligence at the Edge of the Cloud* using the proposed notation. This is the first AI P&ID produced in this notation. The drawing should be read in conjunction with the HAZOP register in the companion paper — every node in that register corresponds to a tagged element on this drawing.

AI-PID-001 · BROADCAST SYNC ARCHITECTURE REV A · 2026 · J. HORVAT



The drawing shows elements that no block diagram of this architecture has previously shown: the sync transmitter ST-001 monitoring the broadcast channel, the sync alarm SA-001 that activates when sync is overdue, the model analyzer MA-001 reading the expert inventory, the coverage indicator CI-001 generating the coverage score signal, and the explicit failure mode annotation on every element. The secure enclave boundary SE-001 is a defined zone, not an implied property.

A process engineer reading this drawing can immediately identify the HAZOP nodes, the instrument loop numbers, and the failure modes. The drawing and the HAZOP register from the companion paper are designed to be used together — every node in that register has a corresponding tagged element on this drawing.

→ [4, 5]

SECTION VI

Version Control and Change Management

A P&ID without version control is not a P&ID. It is a snapshot. The version control and change management

requirements are as important as the notation itself — they are what give the document its legal and operational standing.

In process engineering, every change to a P&ID triggers a Management of Change process. The change is described, the affected nodes are identified, the HAZOP register is updated for affected rows, the change is reviewed by a qualified engineer, and the revision block on the drawing is updated with the revision number, date, description, and responsible engineer. You cannot make an informal change to a P&ID. The document history is part of the document.

CHANGE MANAGEMENT TRIGGERS FOR AI P&ID — PROPOSED REQUIREMENTS

TRIGGER EVENT	REQUIRED ACTION	OWNER
New model version synced — architecture change	Update WM-XXX tag with new version number. Review affected HAZOP rows. Re-check IH version compatibility. Issue new drawing revision.	ML Arch.
New deployment context — physical AI, medical, infrastructure	Full re-HAZOP of affected nodes. Update failure mode annotations for new consequence profile. New drawing revision with context annotation.	Safety Eng.
New external integration added to UI-XXX	Add integration to drawing with data flow type and direction. HAZOP Node 6 re-analysis for new interface. Update privacy boundary annotation.	Platform Eng.
Secure enclave hardware changed	Update SE-XXX annotation with new hardware specification. Re-validate failure mode assumptions for enclave boundary. Security audit triggered.	Security Eng.
Adapter training mechanism changed	Update PA-XXX with new training description. Re-HAZOP Node 3. Update failure mode annotation if drift or over-training behavior changes.	ML Design
Coverage score algorithm changed	Update CI-XXX with new algorithm description. Re-HAZOP Node 5 — coverage overconfidence is the highest-severity open risk. Calibration validation required before release.	ML Design / QA
Production incident with safety consequence	Immediate drawing review. Identify affected tag(s). Determine if drawing was accurate — if not, issue corrective revision. Trigger full HAZOP revalidation for affected nodes.	Safety Eng.

The drawing title block on AI-PID-001 shows revision A. Every subsequent change to the architecture that affects the drawing triggers a new revision — B, C, D — with the date, change description, and responsible engineer recorded. The revision history is permanent. You cannot go back and un-document a change.

This is exactly what process engineering requires of its P&IDs, and for exactly the same reason: when something goes wrong, the first question is always "what did the drawing say, and was the system built to the drawing?" That question requires a drawing that was maintained as a controlled document, not a slide that was last updated eighteen months ago.

→ [2, 3, 6]

SECTION VII

A Path to Standardization

ANSI/ISA-5.1 was first published in 1949 as ISA Recommended Practice RP-5.1. It has been revised multiple times — 1984, 1992, 2009, 2022, 2024 — as the process industries evolved and new technologies required new symbols. The standard did not emerge from a single paper. It emerged from industry consensus over decades, with ISA as the standards body that organized and formalized that consensus.

The same path is available for AI P&ID notation. The symbols and conventions proposed in this paper are a starting point, not a finished standard. What is needed is a standards body willing to take it on, an industry working group willing to develop it, and a critical mass of organizations willing to adopt it.

The most natural home is ISA itself — the International Society of Automation already owns the process P&ID standard and has the expertise and infrastructure to extend it. An AI P&ID standard could be designated ISA-5.5 or ISA-5.X, positioned explicitly as an extension of the existing ISA-5.1 instrumentation and control standards family. The connection to existing standards is a strength — it means process engineers who already use ISA-5.1 can read an AI P&ID with minimal additional training.

Alternatively, IEEE, IEC, or a new AI-specific standards body could adopt the notation. What matters is not which body adopts it but that some body does — that the notation becomes standardized, versioned, and maintained as the technology evolves.

> *ISA-5.1 has been in use for seventy-five years because someone decided that a common notation was worth the effort of standardizing. AI systems will be deployed for at least as long. The time to define the notation is before the accidents, not after.*

→ [1, 2, 7]

REFERENCES

[1] ANSI/ISA-5.1-2024. *Instrumentation and Control — Symbols and Identification.* International Society of Automation. — Current version of the P&ID instrumentation symbol standard; tagging convention adapted in this paper.

[2] ISO 10628-2:2012. *Diagrams for the Chemical and Petrochemical Industry — Graphical Symbols.* International Organization for Standardization. — Process equipment symbol standard; equipment symbol approach adapted for AI elements.

[3] Crawley, F. & Tyler, B. (2015). *HAZOP: Guide to Best Practice.* 3rd ed. Elsevier. — P&ID as prerequisite document for HAZOP; version control and change management requirements.

[4] Horvat, J. (2026). *Intelligence at the Edge of the Cloud: A Local Architecture for the AI Era.* Public domain. — Primary architecture paper; AI-PID-001 documents this architecture.

[5] Horvat, J. (2026). *Applying HAZOP to AI Systems: What Process Engineering Knows About Failure That AI Safety Doesn't.* Public domain. — Companion paper; HAZOP register references tag numbers defined in this drawing.

[6] IEC 61346:1996. *Industrial Systems, Installations and Equipment and Industrial Products — Structuring Principles and Reference Designations.* — Reference designation standard; grounds the tag numbering convention proposed here.

[7] ISA (2024). *ISA Standards and Publications — ISA5.1 Committee.* isa.org. — Standards development process and committee structure for potential adoption of AI P&ID notation as ISA-5.X.

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